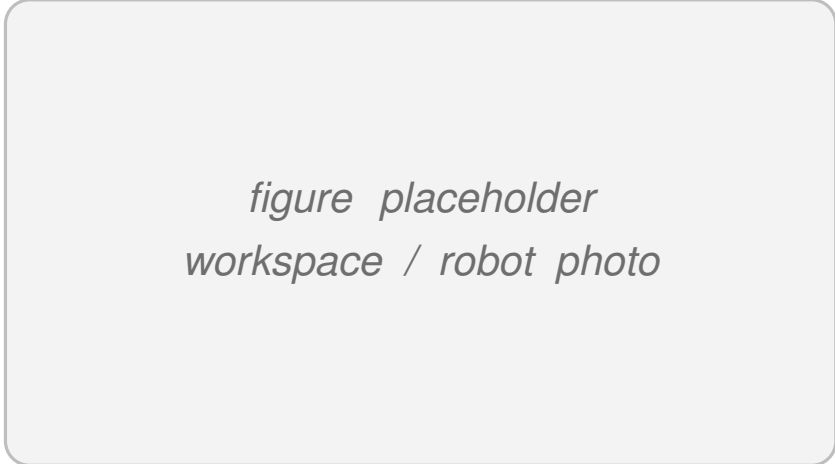


[TODO: one-sentence claim of the poster, first half]
[TODO: second half — keep it to two lines]

Background



[TODO: 2–3 sentences: why a ceiling robot in a home, and why it needs a map of its surroundings rather than a raw point cloud.]

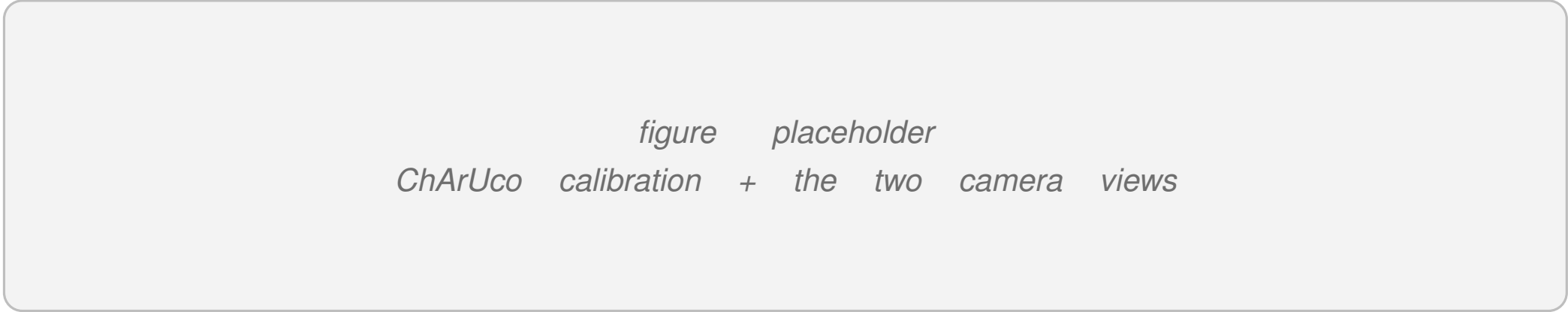
Platform. Two ceiling rails (gantries), two Kinova Gen3 Lite arms each, so every arm is an **8-DOF** chain $\mathbf{q} = [d, \theta, q_1 \dots q_6]$ sharing one moving base with its neighbour. Sensing is **RGB-D only** — no LIDAR.

Problems

- **P1** — [TODO: name the first problem]. [TODO: one or two lines.]
- **P2** — **A raw octomap has no structure.** Occupancy alone gives no topology: no notion of which regions connect, and every frame is paid for again.
- **P3** — **Static and dynamic obstacles are mixed.** With a single live map, the fixed structure of the room and a person walking through it are indistinguishable, so the map is never reproducible between runs.

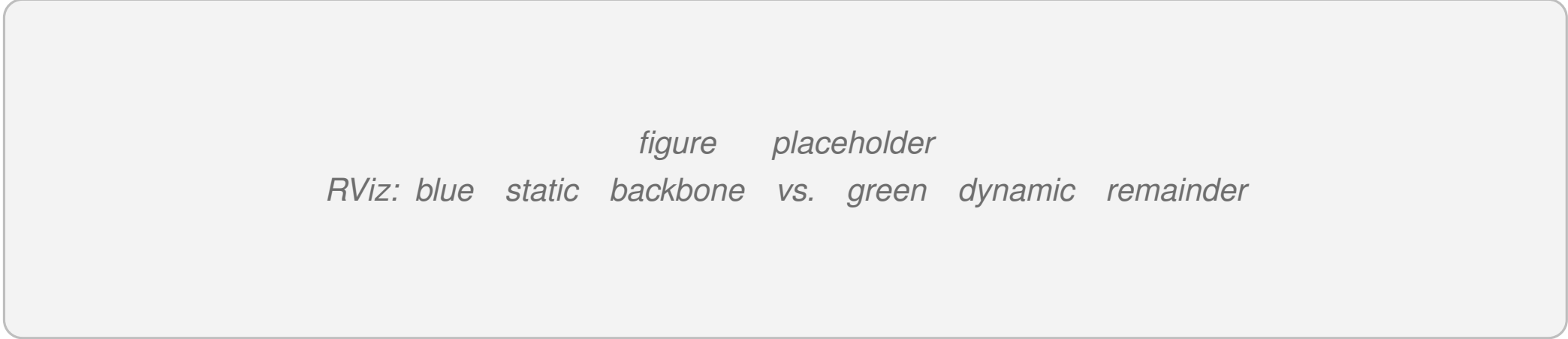
RGB-D Sensing and Extrinsic Calibration

- Two overhead RGB-D cameras are the *only* exteroceptive sensor; their extrinsics are recovered with a **ChArUco** board and published as `world` $\rightarrow$ camera static transforms.
- `depth_cloud` deprojects each depth image into world-frame points **without any segmentation gate**, so the map keeps building even when the detector is blind.
- [TODO: calibration residual / reprojection error, once measured.]

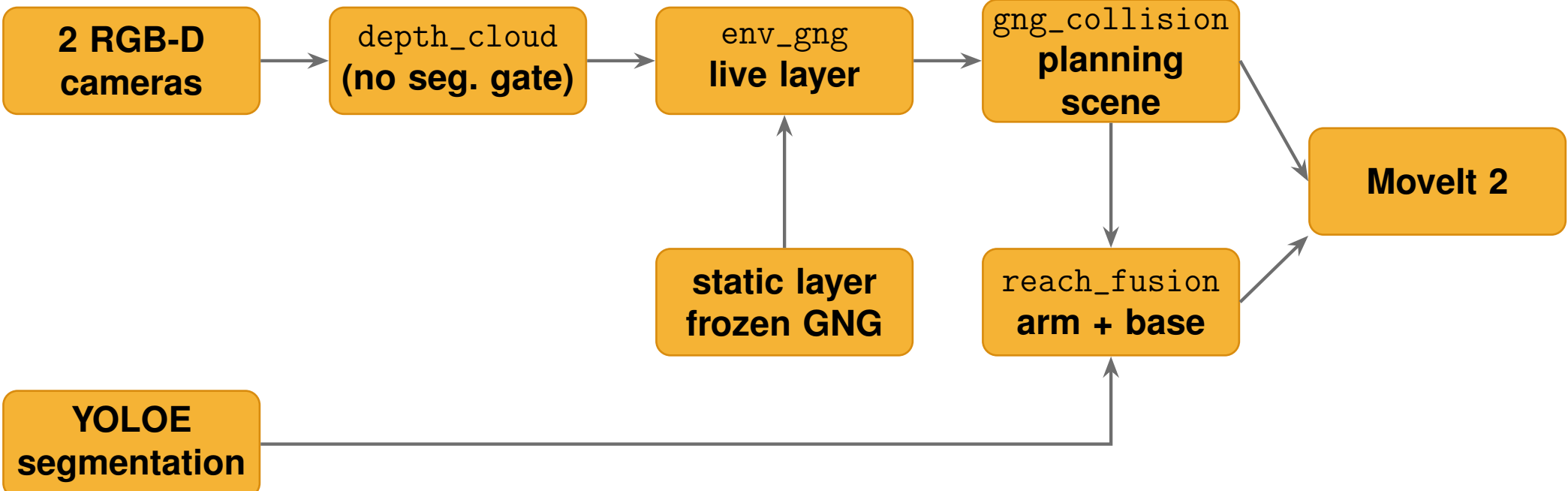


Two-Layer Topological Map (GNG)

- A **Growing Neural Gas** network is fitted to the live depth cloud: nodes tile the observed geometry and competitive Hebbian edges give the scene a *topology*, not just occupancy.
- **Static layer** — captured once on a cleared scene and frozen to a file, giving a backbone that is *identical every run*.
- **Dynamic layer** — live points within `bg_dist` of a static node are subtracted, so the live map carries only the genuinely moving remainder (a person, a displaced object).
- Both layers are unioned into the planning scene, so MoveIt 2 plans against fixed structure and dynamic obstacles at once.



Pipeline



Geometry (top path) is decoupled from identity (bottom path): the topological map builds from `depth_cloud` alone, so it survives a detector that sees nothing.

Object Identity and Tracking

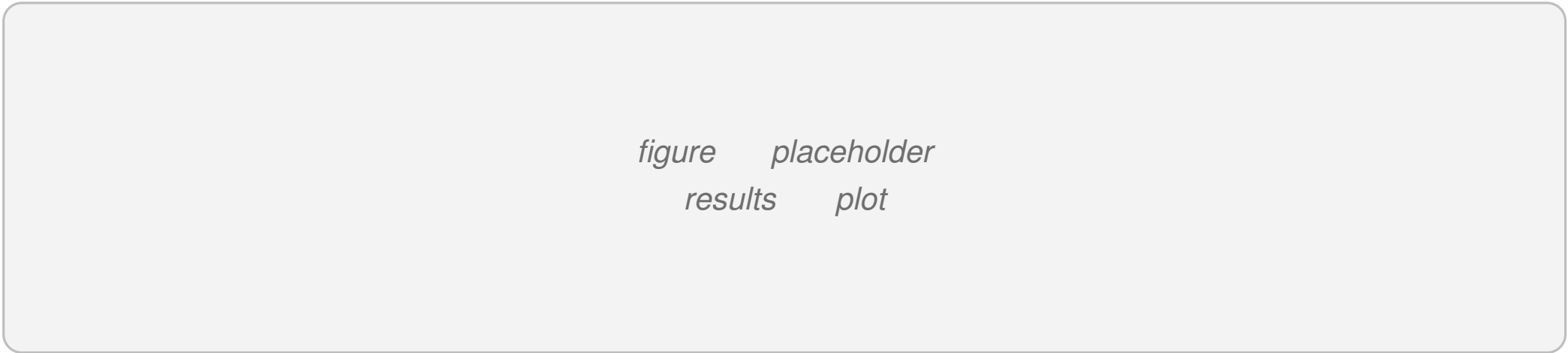
- Detections are associated across frames by **Hungarian matching** with a distance gate and a class-consistency cost, so an object keeps one stable track handle.
- A track is only published after an **N-frame confirmation**, and its class label is decided by **majority voting with hysteresis**, which removes the frame-to-frame label flipping.
- Safety is **geometry-based**: the collision map does not depend on labels, and only the confirmed target is carved out of it.
- [TODO: state which detector the final results use (YOLOE or the simulator's ground-truth segmentation) and why.]

Experiments

E[TODO: n] — [TODO: experiment name]. [TODO: one sentence on the setup: what was varied, how many trials, what was measured.]

Condition	[TODO: metric 1]	[TODO: metric 2]	[TODO: metric 3]
[TODO: baseline]	—	—	—
[TODO: ours]	—	—	—

[TODO: table caption: state the sample size and what "—" will become.]



[TODO: the single headline result, with its number and its honest caveat — write it only once it is measured.]

Summary and Contributions

- [TODO: contribution 1 — the representation.]
- [TODO: contribution 2 — the static/dynamic split.]
- [TODO: contribution 3 — what the evaluation shows.]

Future work. [TODO: one or two sentences.]